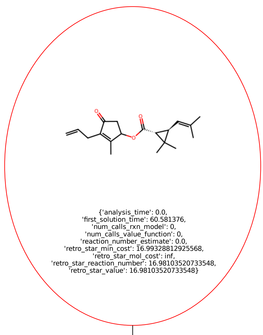
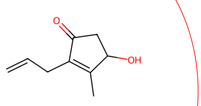
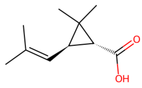


{'analysis_time': 0.0,
'first_solution_time': 60.581376,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 16.993288129255568,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 16.981035207335348,
'retro_star_value': 16.981035207335348}

Reaction
{'analysis_time': 1.747064,
'first_solution_time': 76.696589,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 16.993288129255568,
'retro_star_reaction_number': 16.985255957755842,
'retro_star_rxn_cost': 1.3903023825174294,
'retro_star_value': 16.985255957755842}



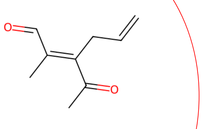
{'analysis_time': 1.747064,
'first_solution_time': 21.307787,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 8.381125019139745,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 8.373092847442428,
'retro_star_value': 16.985255957755842}



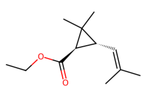
{'analysis_time': 1.747064,
'first_solution_time': 76.696589,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 7.221860727598508,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 7.221860727598508,
'retro_star_value': 16.985255957755842}

Reaction
{'analysis_time': 4.274289,
'first_solution_time': 21.307787,
'num_calls_rxn_model': 21,
'num_calls_value_function': 552,
'retro_star_min_cost': 9.400533823150707,
'retro_star_reaction_number': 8.456361532209093,
'retro_star_rxn_cost': 3.729701448634191 5,
'retro_star_value': 17.068524642325933}

Reaction
{'analysis_time': 2.006416,
'first_solution_time': 76.696589,
'num_calls_rxn_model': 3,
'num_calls_value_function': 79,
'retro_star_min_cost': 7.221860727598508,
'retro_star_reaction_number': 7.221860727598508,
'retro_star_rxn_cost': 0.3552473919475470 6,
'retro_star_value': 16.985255957755842}



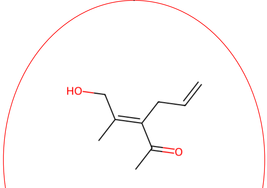
{'analysis_time': 4.274289,
'first_solution_time': 21.307787,
'num_calls_rxn_model': 21,
'num_calls_value_function': 552,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 5.670831374516515,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 4.726660083575801,
'retro_star_value': 17.068524642325933}



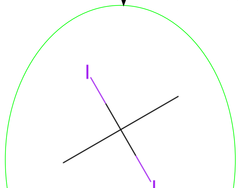
{'analysis_time': 2.006416,
'first_solution_time': 76.696589,
'num_calls_rxn_model': 3,
'num_calls_value_function': 79,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 6.866613335650961,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 6.866613335650961,
'retro_star_value': 16.985255957755842}

Reaction
{'analysis_time': 13.477935,
'first_solution_time': 21.307787,
'num_calls_rxn_model': 84,
'num_calls_value_function': 3046,
'retro_star_min_cost': 5.707489336587527,
'retro_star_reaction_number': 4.726660083575801,
'retro_star_rxn_cost': 0.9969586349416099,
'retro_star_value': 17.068524642325933}

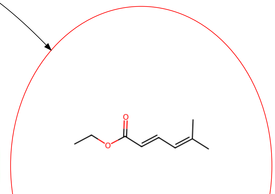
Reaction
{'analysis_time': 2.124912,
'first_solution_time': 76.696589,
'num_calls_rxn_model': 4,
'num_calls_value_function': 105,
'retro_star_min_cost': 6.866613335650961,
'retro_star_reaction_number': 6.866613335650961,
'retro_star_rxn_cost': 6.214600898422191,
'retro_star_value': 16.985255957755842}



{'analysis_time': 13.477935,
'first_solution_time': 21.307787,
'num_calls_rxn_model': 84,
'num_calls_value_function': 3046,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 4.710530701645918,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 3.729701448634191 5,
'retro_star_value': 17.068524642325933}



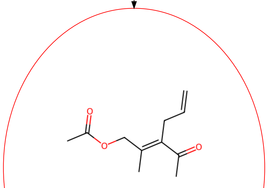
{'analysis_time': 2.124912,
'first_solution_time': 2.124912,
'num_calls_rxn_model': 4,
'num_calls_value_function': 105,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 16.985255957755842}



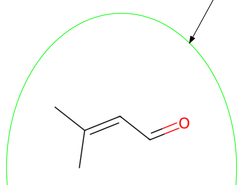
{'analysis_time': 2.124912,
'first_solution_time': 76.696589,
'num_calls_rxn_model': 4,
'num_calls_value_function': 105,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 0.6520052372287701,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 0.6520052372287701,
'retro_star_value': 16.985255957755842}

Reaction
{'analysis_time': 21.307787,
'first_solution_time': 36.291173,
'num_calls_rxn_model': 136,
'num_calls_value_function': 4981,
'retro_star_min_cost': 5.778290526831263,
'retro_star_reaction_number': 5.778290526831263,
'retro_star_rxn_cost': 1.435484005310662 2,
'retro_star_value': 19.117113720523}

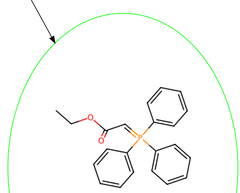
Reaction
{'analysis_time': 76.696589,
'first_solution_time': 76.696589,
'num_calls_rxn_model': 472,
'num_calls_value_function': 23565,
'retro_star_min_cost': 0.6520052372287701,
'retro_star_reaction_number': 0.6520052372287701,
'retro_star_rxn_cost': 0.6520052372287701,
'retro_star_value': 16.985255957755842}



{'analysis_time': 21.307787,
'first_solution_time': 36.291173,
'num_calls_rxn_model': 136,
'num_calls_value_function': 4981,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 4.342805921520600 5,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 4.342805921520600 5,
'retro_star_value': 19.117113720523}

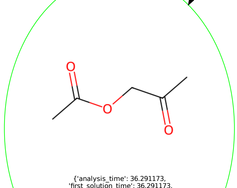


{'analysis_time': 76.696589,
'first_solution_time': 76.696589,
'num_calls_rxn_model': 472,
'num_calls_value_function': 23565,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 16.985255957755842}

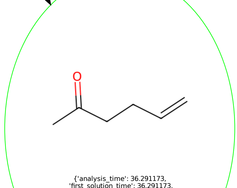


{'analysis_time': 76.696589,
'first_solution_time': 76.696589,
'num_calls_rxn_model': 472,
'num_calls_value_function': 23565,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 16.985255957755842}

Reaction
{'analysis_time': 36.291173,
'first_solution_time': 36.291173,
'num_calls_rxn_model': 234,
'num_calls_value_function': 9483,
'retro_star_min_cost': 4.342805921520600 5,
'retro_star_reaction_number': 4.342805921520600 5,
'retro_star_rxn_cost': 4.342805921520600 5,
'retro_star_value': 19.117113720523}



{'analysis_time': 36.291173,
'first_solution_time': 36.291173,
'num_calls_rxn_model': 234,
'num_calls_value_function': 9483,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 19.117113720523}



{'analysis_time': 36.291173,
'first_solution_time': 36.291173,
'num_calls_rxn_model': 234,
'num_calls_value_function': 9483,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 19.117113720523}